

DINKAR UPADHYAY

dinkarupadhyay2906@gmail.com · [linkedin.com/in/dinkar-up](https://www.linkedin.com/in/dinkar-up) · github.com/dinkar2906

SUMMARY

Backend-focused software engineering student (ECE, graduating Nov 2026) with production internship experience in Java REST API development. GATE qualified in both Computer Science and Data Science & AI. Strong DSA foundation demonstrated through 500+ problems solved across LeetCode and competitive platforms. Hands-on with full-stack AI application delivery — from multimodal ML model training to Android-integrated Flask deployment.

TECHNICAL SKILLS

Languages: C · C++ · Python · Java · SQL

Backend: REST API Design · Flask · Java (Spring concepts) · MySQL

ML / AI: CNN · Multimodal ML · Scikit-learn · Pandas · NumPy

DSA & CP: 500+ problems solved (LeetCode + other platforms) · Data Structures · Algorithms

Tools & OS: Git · GitHub · Linux · Android Studio · Power BI

EXPERIENCE

Software Engineering Intern — Singsys Pte Ltd *Lucknow, India · Jun 2025 – Jul 2025*

Java · REST APIs · Agile

- Built and maintained production-grade RESTful API endpoints in Java, contributing directly to live backend services handling real client traffic.
- Identified and resolved code-quality issues during peer code reviews, leading to cleaner, more maintainable service modules.
- Collaborated with senior engineers across the full software delivery lifecycle — from architecture discussions to deployment — gaining end-to-end system ownership exposure.

Technical Support Engineer — Infobay AI (EduGorilla) *Lucknow, India · Apr 2023 – Dec 2024*

SaaS · Bug Reproduction · Cross-functional Collaboration

- Diagnosed and resolved client-reported bugs in SaaS products by reproducing edge-case issues in staging and coordinating directly with the development team to ship fixes.
- Wrote structured bug reports and validated patches before release, reducing average bug-to-fix cycle time for recurring issue categories.
- Authored internal troubleshooting runbooks that became the team's standard reference, measurably cutting average issue-resolution time across the support function.

PROJECTS

AI-Based Stress Detection System · *Python · Flask · Android · CNN · Multimodal ML*

- Designed and trained a multimodal deep-learning pipeline combining facial expression, voice, and ECG (CNN) signals to classify stress levels — achieving strong accuracy across all three modalities.
- Exposed inference logic via a Flask REST backend and integrated it with a native Android client for real-time stress analysis and result visualisation, enabling end-to-end user-facing delivery.
- Managed the complete ML lifecycle independently: data preprocessing, model training, evaluation, and local server deployment.

Ray Tracer · *C++ · Physically-Based Rendering · Systems Programming*

- Implemented a physically-based ray tracer from scratch in C++, simulating reflection, refraction, shading, and geometric transformations without relying on any graphics library.
- Optimised ray-scene intersection logic and memory layout, demonstrating strong command of systems programming fundamentals and computational geometry.

EDUCATION

CERTIFICATIONS & ACHIEVEMENTS

- **GATE 2x Qualified:** GATE Qualified — Data Science & AI (GATE DA) and Computer Science Engineering (GATE CSE).
- **IBM (Honors):** IBM Professional Certifications (Coursera, May 2024, with Honors) — Python for Data Science, Data Analysis with Python, Machine Learning with Python, Databases & SQL for Data Science.
- **Microsoft/LinkedIn:** Microsoft & LinkedIn — Career Essentials in Data Analysis (Jul 2024) — Applied data analysis and BI using Excel and Power BI.
- **DSA Practice:** 500+ problems solved across LeetCode and other competitive programming platforms, with consistent practice in arrays, graphs, DP, and system design patterns.